

# Lecture Homework Week 06 - Thursday

## Template

### Student Information

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### Student Responses

Answer the following questions

#### Question 1

When running the Keyboard Demo in QEMU after introducing the bug, briefly describe how it behaves differently than before. Try a few keystrokes in the QEMU window and describe what you see:

Before fixing the code, it was stuck in a “Timer with delta zero, disabling” infinite loop on the terminal. Entering keystroke on the QEMU make the text appear where its not suppose to like right next to the red text.

#### Question 2

What do you think is happening? The program continues to execute, but what happens after it returns from `irq_handler`? What C subroutine gets called over and over?

Because we aren't restoring everything, when it gets called again, it overwrite the data and corrupts it since `irq_handler` gets called repeatedly.

### Question 3

Look at the terminal where the buggy version is running, what messages are being printed out about the timers? Do the timers work in this buggy version?

“Timer with delta zero, disabling.” The timer aren’t fully working because the terminal get stuck in a infinite loop, displaying that same message.